MAT627 Chapter 2.4 Programming Assignment

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We approximate $f(x) = x^{1/3}$ by uniform piecewise linear interpolation and measure the error on the fine grid

$$D = \{z_i\}_{i=0}^{1000}, \qquad z_i = a + i\frac{b-a}{1000}.$$

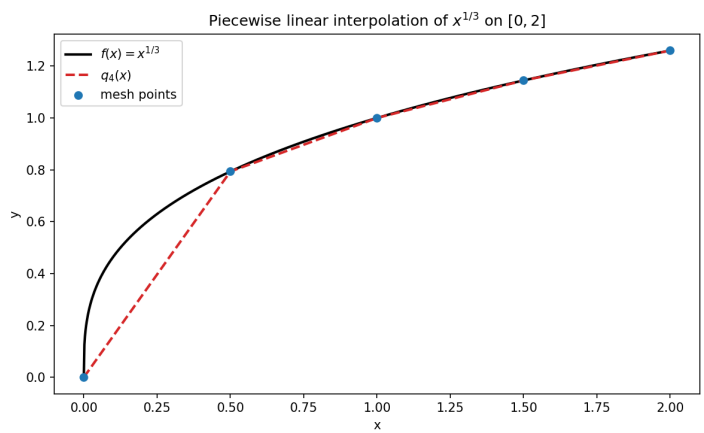

Figure 1: Part 1: $q_4(x)$ and $f(x) = x^{1/3}$ on $[0, 2]$.

1 Part 2

Uniform piecewise linear interpolation on $[1, 2]$.

2 Part 3

Uniform piecewise linear interpolation on $[0, 1]$.

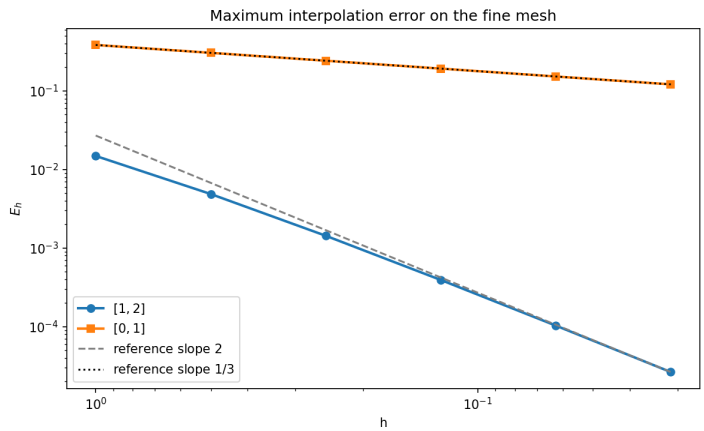

Figure 2: Error decay on the fine grid for the intervals $[1, 2]$ and $[0, 1]$.

Table 1: Maximum fine-grid error for $f(x) = x^{1/3}$ on $[1, 2]$.

N	h	$E_h = \max_{z_i \in D} \lvert q_N(z_i) - f(z_i) \rvert$	Rate
1	1	1.488703×10^{-2}	–
2	0.5	4.875503×10^{-3}	1.610433
4	0.25	1.434610×10^{-3}	1.764892
8	0.125	3.929018×10^{-4}	1.868418
16	0.0625	1.031247×10^{-4}	1.929779
32	0.03125	2.641642×10^{-5}	1.964883

Table 2: Maximum fine-grid error for $f(x) = x^{1/3}$ on $[0, 1]$.

N	h	$E_h = \max_{z_i \in D} \lvert q_N(z_i) - f(z_i) \rvert$	Rate
1	1	3.848998×10^{-1}	–
2	0.5	3.054952×10^{-1}	0.333333
4	0.25	2.424717×10^{-1}	0.333333
8	0.125	1.924499×10^{-1}	0.333333
16	0.0625	1.527476×10^{-1}	0.333333
32	0.03125	1.212358×10^{-1}	0.333333

3 Discussion

On $[1, 2]$, the second derivative

$$f''(x) = -\frac{2}{9x^{5/3}}$$

is bounded, so the standard piecewise linear interpolation estimate applies and the observed rate is approximately 2.

On $[0, 1]$, the second derivative is singular at $x = 0$, so the theorem does not apply. On the first cell $[0, h]$, writing $x = th$ with $0 \leq t \leq 1$, the interpolant is

$$q_N(x) = th^{1/3},$$

which gives

$$|f(x) - q_N(x)| = h^{1/3}|t^{1/3} - t|.$$

Therefore the dominant error behaves like $O(h^{1/3})$, so the rate is not second order near the origin and instead approaches $1/3$.